

Synopsis

Design and Development of an End-to-End Machine Learning System to Predict User Movie Preferences Based on Netflix User Rating History

Problem Id: 17

Team Leader:

Student Name: Krish Verma

University Roll number: 2028748

Section: L

Team Member:

Student Name: Dev Dhaka

University Roll number: 2028699

Section: L

Team Member:

Student Name: Dhiraj Shah

University Roll number: 2028709

Section: L

1.PROBLEM DESCRIPTION:

Movie streaming services like Netflix generate a lot of user rating information. Using machine learning to predict user preferences for movies can improve the recommendation system. The project's goal is to create a machine learning system that predicts whether a user likes or dislikes a movie based on their past ratings. The dataset for the project includes user ID, movie ID, ratings, and timestamps. The machine learning system cleans the data by addressing missing values, removing duplicate records, and standardizing the ratings. Key features, such as viewing patterns and the movie's popularity, will be included. The project will use machine learning algorithms like Logistic Regression, K-Nearest Neighbors, and Support Vector Machine to classify movie preferences as recommended or not recommended. The performance of these algorithms will be evaluated using accuracy, precision, recall, F1 score, confusion matrix, and ROC curve. This system will help improve the recommendation systems used in real-world streaming services.

STEPS OF IMPLEMENTATION:

1. Data collection A suitable dataset, like the Netflix Prize dataset or the MovieLens dataset that contains user ratings, will be selected.

	userId	movieId	rating	timestamp
0	196	242	NaN	881250949.0
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1	186	302	3.0	891717742.0
2	22	377	1.0	878887116.0
3	244	51	2.0	880606923.0
4	166	346	1.0	886397596.0

2. Data preprocessing The dataset will be cleaned by removing missing values and duplicate records. Ratings will be normalized, and categorical variables like user ID and movie ID will be encoded.

Processed Data							
	userId	movieId	rating	timestamp	label	movie_popularity	user_avg_rating
1	161	257	3.0	891717742.0	0	7	3.666667
2	20	311	1.0	878887116.0	0	1	2.000000
3	216	41	2.0	880606923.0	0	3	3.625000
4	142	291	1.0	886397596.0	0	5	3.000000
5	267	387	4.0	884182806.0	1	4	3.615385

3. Feature engineering We will extract important features such as user rating behavior, movie popularity, and historical rating patterns from the dataset.
4. Feature selection we will choose relevant features using statistical or correlation techniques to improve model performance.

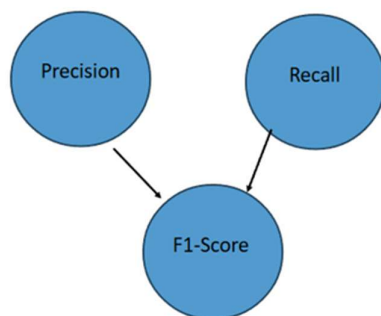
5. Model training three machine learning algorithms will be used:
 - Logistic Regression
 - K-Nearest Neighbors
 - Support Vector Machine
6. Model evaluation We will evaluate the trained models using classification performance metrics.
7. Prediction system The final system will predict whether a movie will receive a high rating (recommended) or a low rating (not recommended) from a user.

EXPECTED MODULES, LIBRARIES TO BE USED:

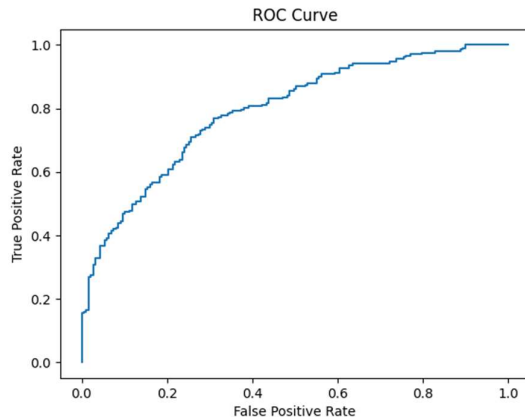
1. NumPy used for numerical operations and handling large arrays.
2. Pandas used for data manipulation, cleaning, and preprocessing.
3. Matplotlib used for visualizing graphs such as ROC curves and plots.
4. Seaborn used for data visualization and correlation analysis.
5. sklearn provides machine learning algorithms like Logistic Regression, KNN, and SVM, along with evaluation metrics.

EXPECTED EVALUATION METRICS AND PLOTS TO BE USED:

1. Accuracy measures the percentage of correctly predicted movie preferences.
2. Precision shows how many predicted positive recommendations are correct.
3. Recall measures how well the model identifies all relevant recommendations.
4. F1-score balances precision and recall.



5. Confusion matrix shows the number of correct and incorrect predictions made by the model.
6. ROC curve evaluates the classification performance of the machine learning



MODELS REQUIRED TOPICS FROM THE SUBJECT:

1. Strings
Strings can be used to handle text-based data such as movie titles and genres in the data set.
2. Pointers
Pointers can help in efficient memory management and data handling in algorithm implementation.
3. Files
File handling can be used to handle data sets (CSV files) store processed data, and results

PLATFORM REQUIRED:

Python

Jupyter Notebook

Visual Studio Code / Python IDE Books and Link Sources:Books and Link

SOURCES:

Tom Mitchell – *Machine Learning*

<https://scikit-learn.org> (Machine Learning documentation)

<https://www.kaggle.com> (Dataset source)

<https://scikit-learn.org>

<https://pandas.pydata.org>